Aga Khan University Examination Board

Notes from E-Marking Centre HSSC-II Computer Science Annual Examinations 2023

Introduction

This document has been prepared for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part II (Class XII) Computer Science. It contains comments on candidates' responses to the 2023 HSSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

Most candidates achieved success in constructing good responses specifically in the following topics.

- Operating System
- System Development Life Cycle (SDLC)
- Introduction to C++ Programming
- Control Structures

Nonetheless, it is essential for teachers to concentrate on the following concepts and provide candidates with more practice to foster a solid understanding.

- Arrays and Strings
- Functions
- Object Oriented Programming (OOP)
- File Handling

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

Detailed Comments

Constructed Response Questions (CRQs)

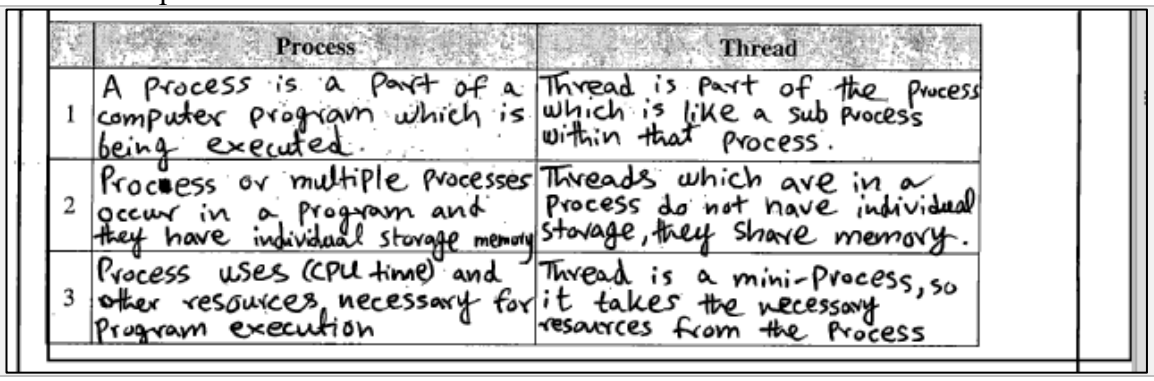
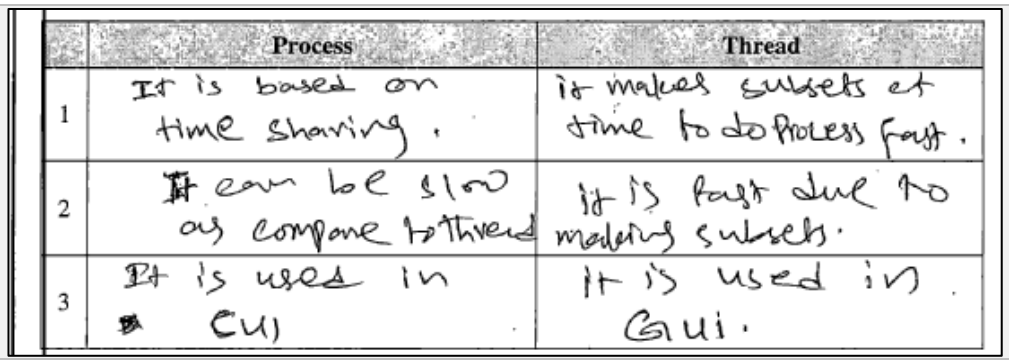
<table>
<tr><th colspan="2">Question No. 1</th></tr>
<tr><td>Question Text</td><td>Differentiate between process and thread.
<table>
<tr><th></th><th>Process</th><th>Thread</th></tr>
<tr><td>1</td><td></td><td></td></tr>
<tr><td>2</td><td></td><td></td></tr>
<tr><td>3</td><td></td><td></td></tr>
</table></td></tr>
<tr><td>SLO No.</td><td>9.3.3</td></tr>
<tr><td>SLO Text</td><td>Differentiate between thread and process.</td></tr>
<tr><td>Max Marks</td><td>3</td></tr>
<tr><td>Cognitive Level</td><td>U*</td></tr>
<tr><td>Checking Hints</td><td>1 mark for writing each difference. (Any THREE required)
No mark will be awarded for writing incomplete difference.</td></tr>
<tr><td>Overall Performance</td><td>The cohort’s overall performance in this question was of average standard. Some candidates encountered difficulty in accurately distinguishing between processes and threads. However, several candidates exhibited a clear understanding of the topic by providing correct responses.</td></tr>
<tr><td>Description of Better Responses</td><td>Better responses correctly showed the fundamental differences between process and thread. The presentation of differences demonstrated comprehensive understanding. The responses effectively highlighted that a process pertains to the execution of a computer program, while a thread constitutes a sub-process within it. Candidates wrote correct difference in terms of memory and resource usage which shows the clear understanding of the concept.</td></tr>
<tr><td>Image of Better Response</td><td> 
<table>
<tr><th></th><th>Process</th><th>Thread</th></tr>
<tr><td>1</td><td>A process is a part of a computer program which is being executed.</td><td>Thread is part of the process which is like a sub process within that process.</td></tr>
<tr><td>2</td><td>Process or multiple processes occur in a program and they have individual storage memory</td><td>Threads which are in a process do not have individual storage, they share memory.</td></tr>
<tr><td>3</td><td>Process uses (CPU time) and other resources necessary for Program execution</td><td>Thread is a mini-Process, so it takes the necessary resources from the Process</td></tr>
</table></td></tr>
<tr><td>Description of Weaker Responses</td><td>Weaker responses were less effective and contained extra information, deviating from the topic. The points presented lack of relevance and use of inappropriate vocabulary, resulting in an inaccurate representation of the key definitions of process and thread. Mostly candidates mixed up the concept of memory and resource sharing with time sharing, which was incorrect. It is essential for the candidates to focus on pertinent content and employ accurate terminologies to convey the precise distinctions between these concepts.</td></tr>
</table>

Image of Weaker Response	



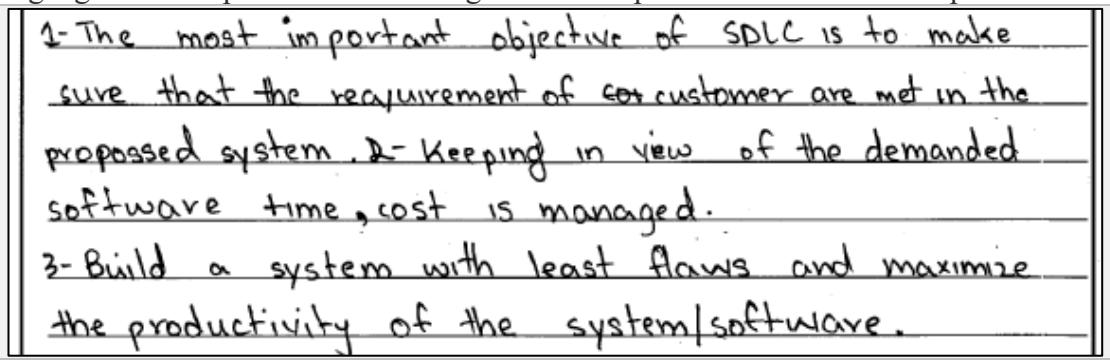
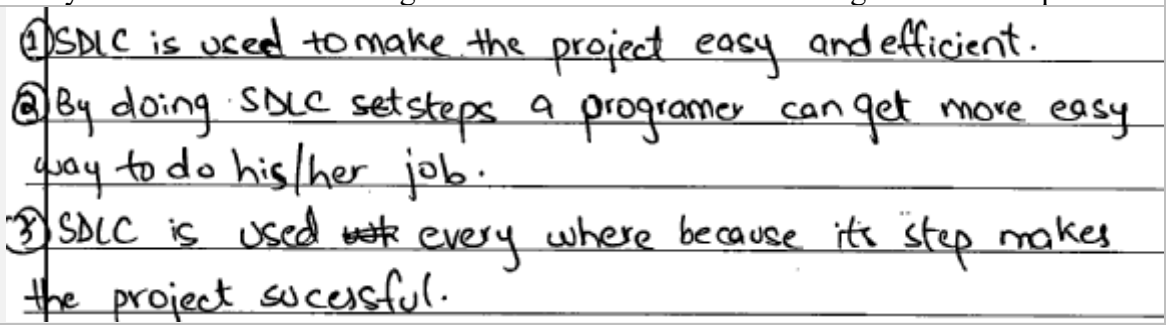
	Process	Thread
1	It is based on time sharing.	it makes subsets of time to do process fast.
2	It can be slow as compare to thread	it is fast due to making subsets.
3	It is used in CUI	it is used in Gui.

Suggestions for improvement (Highlighted part)

How to Approach SLO	Pedagogy** Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept mapping • Audio Visual resources • Think, pair and share • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform
Any Additional Suggestion: Teachers are advised to use simulations and presentations such as https://www.youtube.com/watch?v=4rLW7zg21gI to explain the topic in detail.		

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 2	
Question Text	Write THREE main objectives of System Development Life Cycle (SDLC).
SLO No.	10.1.2
SLO Text	Describe objectives of SDLC.
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for writing description of each state. (THREE required)
Overall Performance	The cohort showed exceptional performance in addressing the question about the main goal of the System Development Life Cycle (SDLC). However, some encountered challenges in articulating the essential points effectively. While many candidates demonstrated an understanding of the concept, refining the ability to concisely convey key aspects is encouraged.
Description of Better Responses	Such responses clearly outlined the primary objective of SDLC, highlighting the value of conceptual clarity. Candidates effectively conveyed the points like cost and time management, requirement specification and maximise the productivity of the software are the objectives of

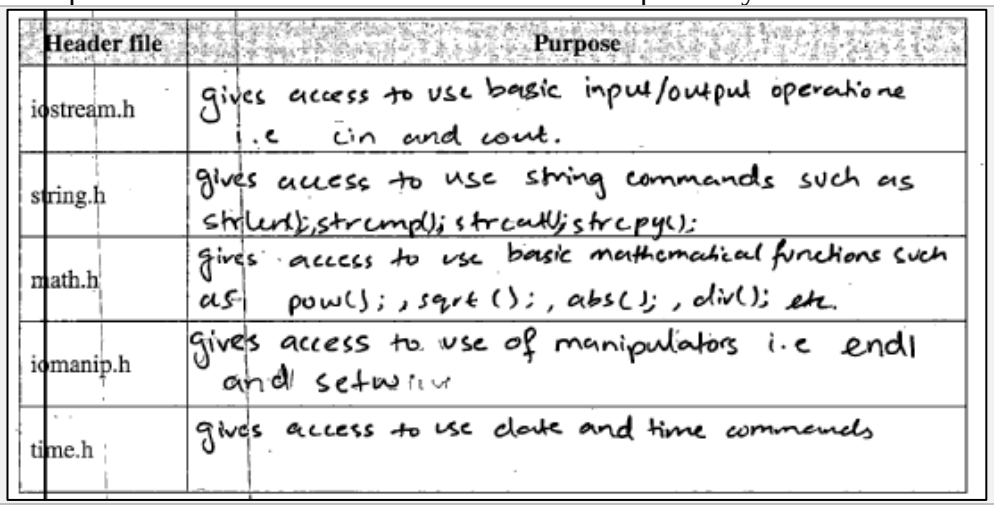
	SDLC. Most of the responses showed that the candidates understood the topic well and highlighted the importance of following a structured process in software development.
Image of Better Response	
Description of Weaker Responses	Weaker responses showed shortcomings in important information and included inaccuracies. The correct objective of SDLC was not defined accurately, demonstrating lack of comprehension. Some candidates wrote that it helps the programmer to do the job easily. Some candidates wrote general answers which were not aligned with the question.
Image of Weaker Response	

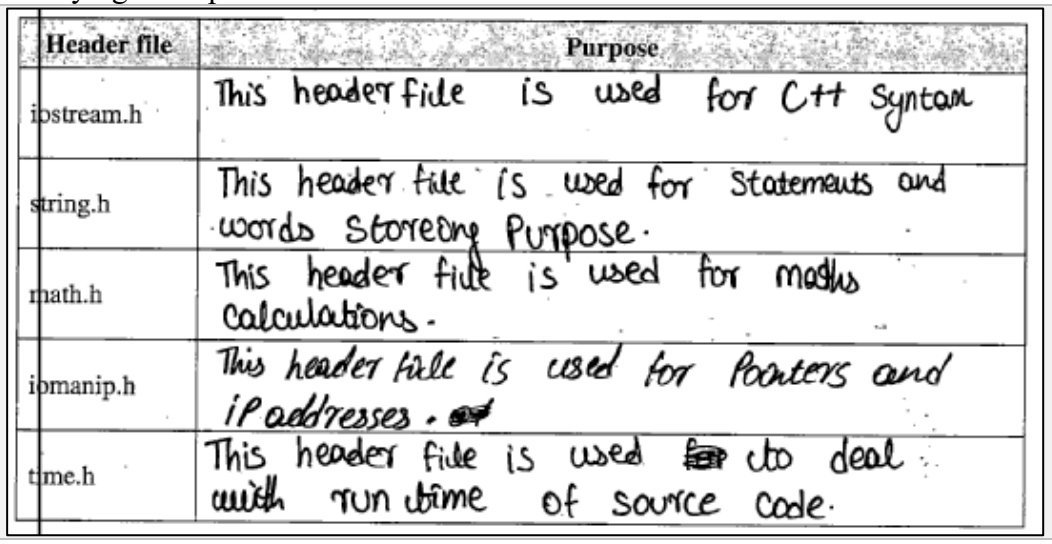
Suggestions for improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept mapping • Audio Visual resources • Think, pair and share • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform

Any Additional Suggestion:
Teachers are advised to use video resources such as https://www.youtube.com/watch?v=-qlCj9Jt8vQ for better understanding of the topic.

<table>
<tr><th colspan="2">Question No. 3</th></tr>
<tr><td>Question Text</td><td>Describe the purpose of each of the given header files.

<table><tr><th>Header file</th><th>Purpose</th></tr><tr><td>iostream.h</td><td></td></tr><tr><td>string.h</td><td></td></tr><tr><td>math.h</td><td></td></tr><tr><td>iomanip.h</td><td></td></tr><tr><td>time.h</td><td></td></tr></table></td></tr>
<tr><td>SLO No.</td><td>11.1.2</td></tr>
<tr><td>SLO Text</td><td>Explain the basic structure of a basic C++ program including pre-processor directives, main function, body of program.</td></tr>
<tr><td>Max Marks</td><td>5</td></tr>
<tr><td>Cognitive Level</td><td>U</td></tr>
<tr><td>Checking Hints</td><td>1 mark for writing purpose of each header file. (FIVE required)</td></tr>
<tr><td>Overall Performance</td><td>A significant number of candidates provided correct responses and showed a solid comprehension of the purpose of header files in the C++ language. They demonstrated a good understanding of how header files are used to include function, definitions, and declarations in a program. However, there were a few candidates who had some areas that required improvement. Such candidates provided incomplete or inaccurate explanations or lacked a clear understanding of the role of given header files in C++ programming.</td></tr>
<tr><td>Description of Better Responses</td><td>Better responses correctly described the precise functions of distinct header files in the C++ language, supported by examples that showcased a meticulous comprehension of the concepts specifically using math.h, iomanip.h and time.h for builtin math functions, manipulator functions and date/time functions respectively.</td></tr>
<tr><td>Image of Better Response</td><td> 
<table>
<tr><th>Header file</th><th>Purpose</th></tr>
<tr><td>iostream.h</td><td>gives access to use basic input/output operations i.e cin and cout.</td></tr>
<tr><td>string.h</td><td>gives access to use string commands such as strlen();strcmp();strcat();strcpy();</td></tr>
<tr><td>math.h</td><td>gives access to use basic mathematical functions such as pow();, sqrt();, abs();, div(); etc.</td></tr>
<tr><td>iomanip.h</td><td>gives access to use of manipulators i.e endl and setw();</td></tr>
<tr><td>time.h</td><td>gives access to use date and time commands</td></tr>
</table></td></tr>
</table>

Description of Weaker Responses	Some candidates' responses were less effective due to inaccuracies in defining the purpose of header files specifically they wrote wrong description of iomanip.h, math.h, and time.h file. The explanation lacked required conceptual depth and accuracy, failing to sufficiently clarifying the topic.
Image of Weaker Response	

Header file	Purpose
iostream.h	This header file is used for C++ Syntax
string.h	This header file is used for Statements and words Storeing Purpose.
math.h	This header file is used for maths calculations.
iomanip.h	This header file is used for Pointers and IP addresses.
time.h	This header file is used to deal with run time of source code.

Suggestions for improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept mapping • Audio Visual resources • Think, pair and share • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform

Any Additional Suggestion:
Teachers are advised to give more practice problems that use different header files such as calculating square root of a number using math.h file.

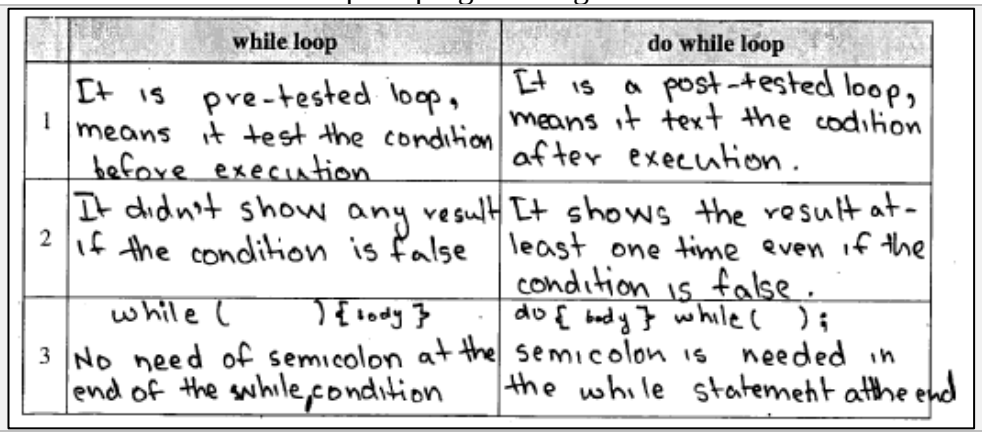
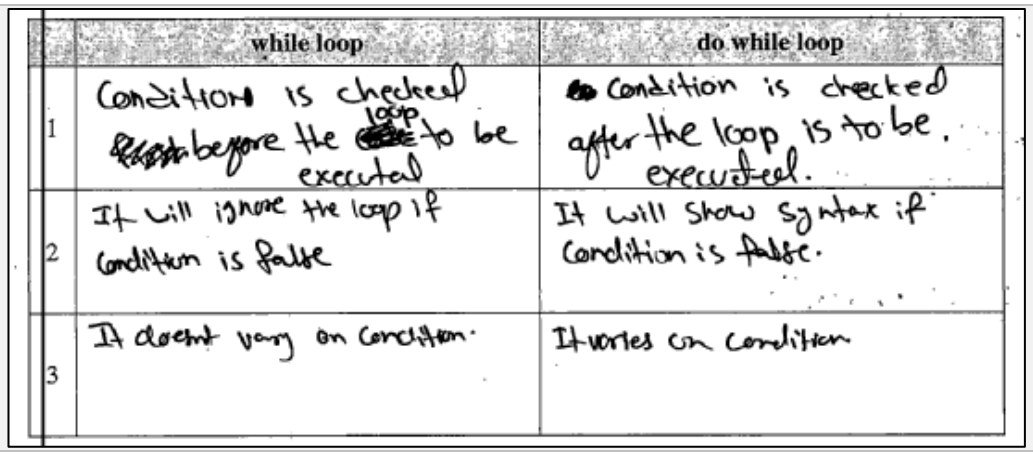
<table>
<tr><th colspan="2">Question No. 4</th></tr>
<tr><td>Question Text</td><td>Differentiate between while loop and do while loop statement.
<table>
<tr><th></th><th>while loop</th><th>do while loop</th></tr>
<tr><td>1</td><td></td><td></td></tr>
<tr><td>2</td><td></td><td></td></tr>
<tr><td>3</td><td></td><td></td></tr>
</table></td></tr>
<tr><td>SLO No.</td><td>12.2.5</td></tr>
<tr><td>SLO Text</td><td>Differentiate between:
a. for and while loop
b. while and do while loop</td></tr>
<tr><td>Max Marks</td><td>3</td></tr>
<tr><td>Cognitive Level</td><td>U</td></tr>
<tr><td>Checking Hints</td><td>1 mark for writing each correct difference. (Any THREE required)
No mark will be awarded on writing incomplete difference.</td></tr>
<tr><td>Overall Performance</td><td>The performance of the cohort in this question can be classified as average. Several candidates were not able to incorporate essential elements expected in their responses. While certain answers effectively showed relevant concepts, a portion of candidates overlooked key aspects, leading to their responses not meeting higher-mark criteria.</td></tr>
<tr><td>Description of Better Responses</td><td>The candidates' responses clearly showed the principles of the while loop and do while loop, highlighting their functionalities and distinctions as pre-tested and post-tested loops with the least number of iterations performed by each loop on identifying wrong condition. Furthermore, the inclusion of syntax with relevant parameters enhanced clarity in understanding these loop structures. Their comprehensive approach showcased a commendable grasp of the topic, allowing for a deeper insight into the utilisation and differentiation of these loops in programming.</td></tr>
<tr><td>Image of Better Response</td><td> 
<table>
<tr><th></th><th>while loop</th><th>do while loop</th></tr>
<tr><td>1</td><td>It is pre-tested loop, means it test the condition before execution</td><td>It is a post-tested loop, means it text the codition after execution.</td></tr>
<tr><td>2</td><td>It didn't show any result if the condition is false</td><td>It shows the result at-least one time even if the condition is false.</td></tr>
<tr><td>3</td><td>while ( ) {body} No need of semicolon at the end of the while condition</td><td>do {body} while( ); semicolon is needed in the while statement at the end</td></tr>
</table></td></tr>
<tr><td>Description of Weaker Responses</td><td>The weaker responses indicated a limited grasp of do while and while loop concepts, incorporating inaccurate and irrelevant elements. Some candidates only wrote about the conditions checked in the loop resulting in losing marks. This suggested a deficiency in comprehending the purpose and operational principles of these loop structures.</td></tr>
</table>

Image of Weaker Response	

	while loop	do while loop
1	Condition is checked before the loop to be executed	Condition is checked after the loop is to be executed.
2	It will ignore the loop if condition is false	It will show syntax if condition is false.
3	It doesn't vary on condition.	It varies on condition.

Suggestions for improvement (Highlighted part)

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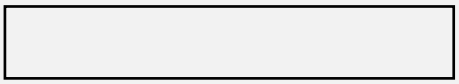
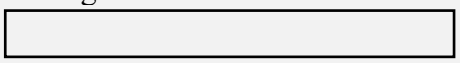
Any Additional Suggestion:
Teachers are advised to give more practice problems of loops and ask them to debug the program by applying breakpoint.

Question No. 5

Question Text

Complete the given C++ code to create an object and call the member function using object oriented programming.

```
#include <iostream>
using namespace std;
class Animal {
  public:
   void eat() {
      cout << "I can eat!" << endl;
   }
   void sleep() {
      cout << "I can sleep!" << endl;
   }
};
class Dog : public Animal {
  public:
   void bark() {
      cout << "I can bark! Woof woof!!" << endl;
```

	}
	};
	int main() {
	
	// Create object of the Dog class
	 
	
	// Calling members of the base class
	 
	
	// Calling member of the derived class
	 
	
	return 0;
	}

SLO No.	16.3.7
SLO Text	Write the syntax of inheritance for three access specifiers.
Max Marks	4
Cognitive Level	*A

Checking Hints	1 mark for creating object of dog class
	Dog dog1;
	1 mark for calling members of the base class (2 required)
	dog1.eat();
	dog1.sleep();
	1 mark for calling member of the derived class
	dog1.bark();

Overall Performance	This question pertained to the establishment of classes and objects within the C++ language. Responses earning full marks proficiently illustrated accurate creation of class objects and the calling of member functions from both base and derived classes. Marks were allocated solely for correctly calling member function. The cohort's collective performance in this domain was average, showcasing a range of outcomes. Further focus on meticulous application of class-object concepts and accurate member function usage could enhance overall proficiency in similar scenarios.
Description of Better Responses	Better responses were noteworthy, showcasing a precise and accurate comprehension of class and object creation. The inclusion of correct syntax for generating class (dog) object and invoking member functions (eat, sleep and bark) from the classes demonstrated a robust understanding of the subject matter.

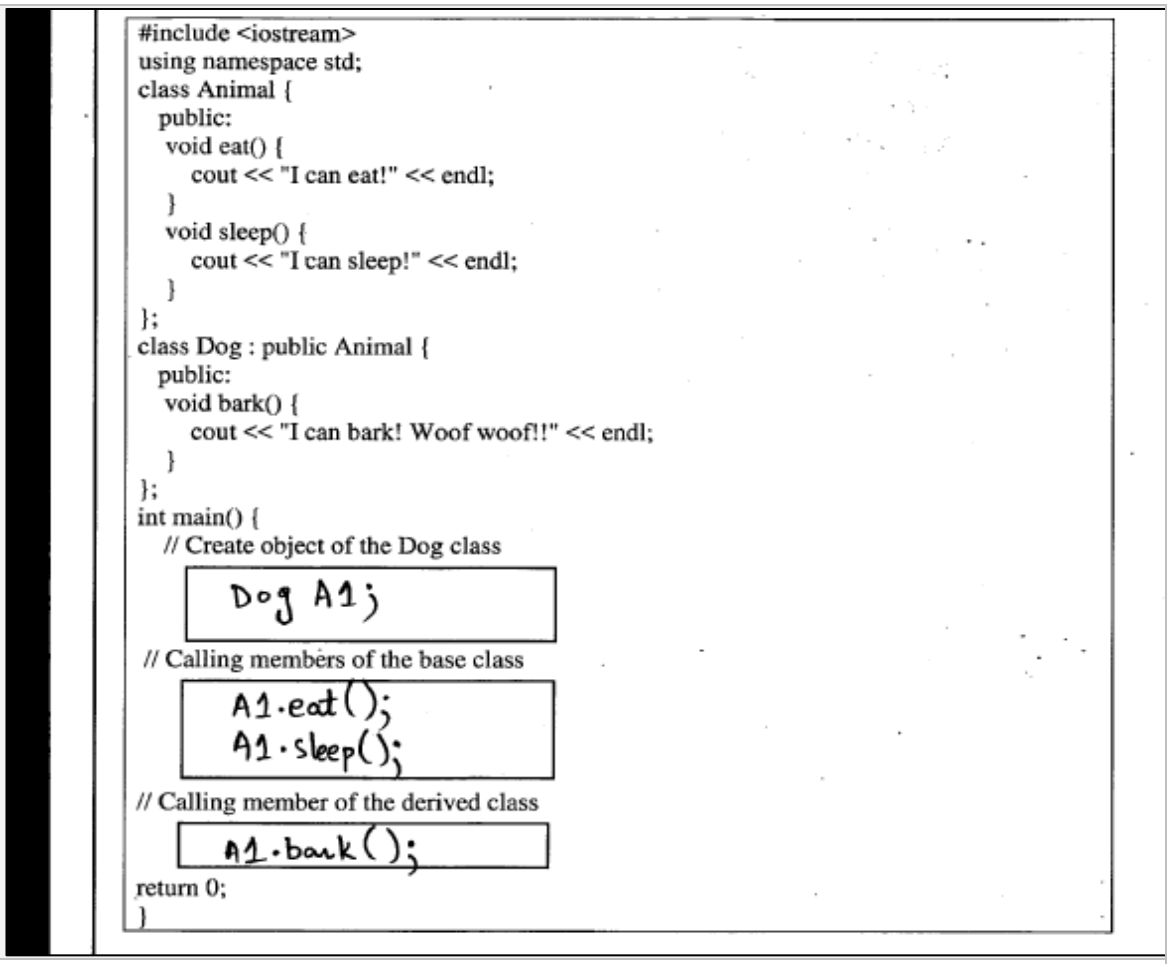
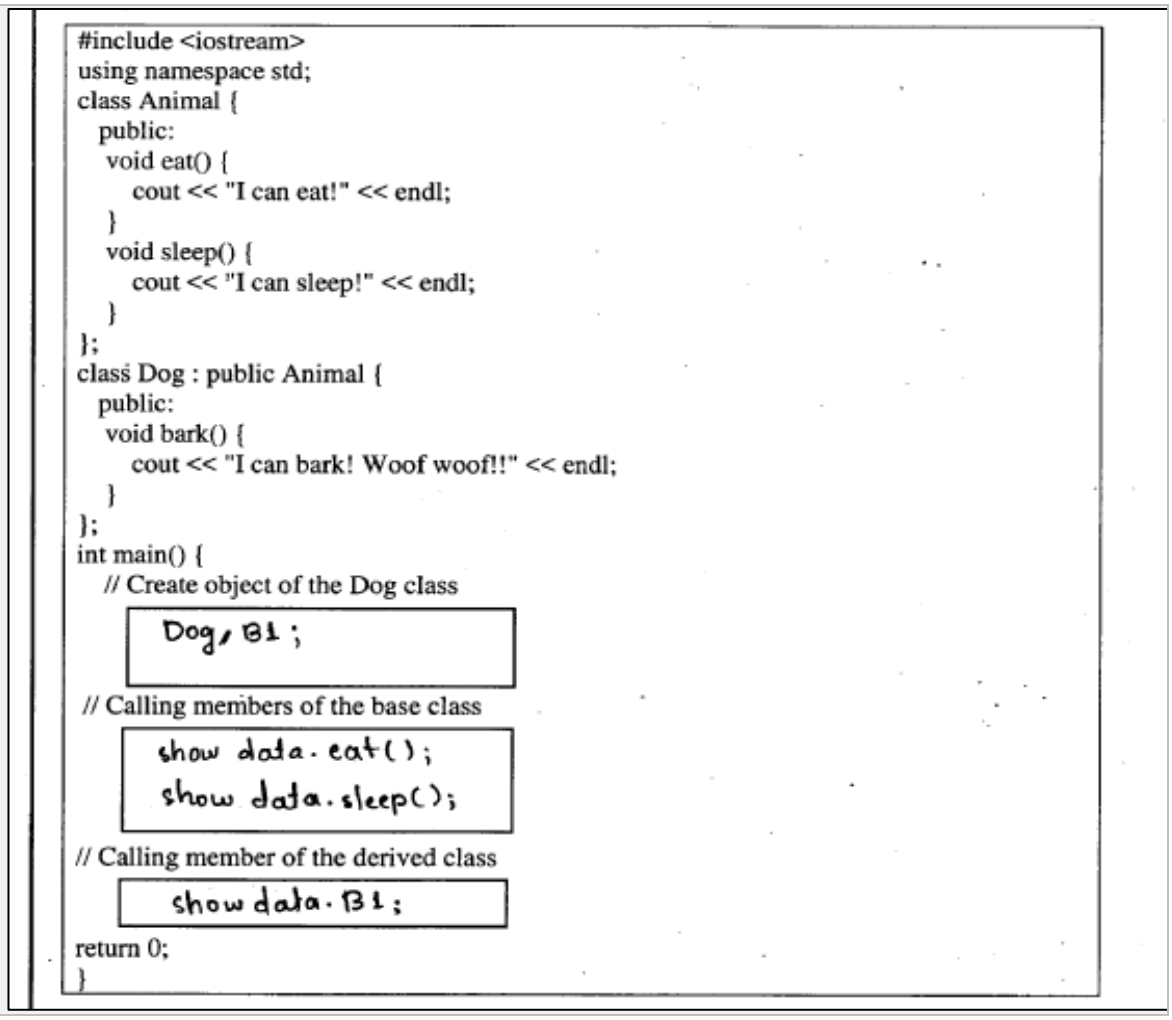
Image of Better Response	
Description of Weaker Responses	Weaker responses demonstrated shortcomings, encompassing erroneous declarations and method invocations. Some candidates used incorrect syntax that reflected a limited grasp of fundamental concepts encompassing class creation, object declaration, and calling method. There was a notable absence of clarity and comprehension regarding these essential aspects.

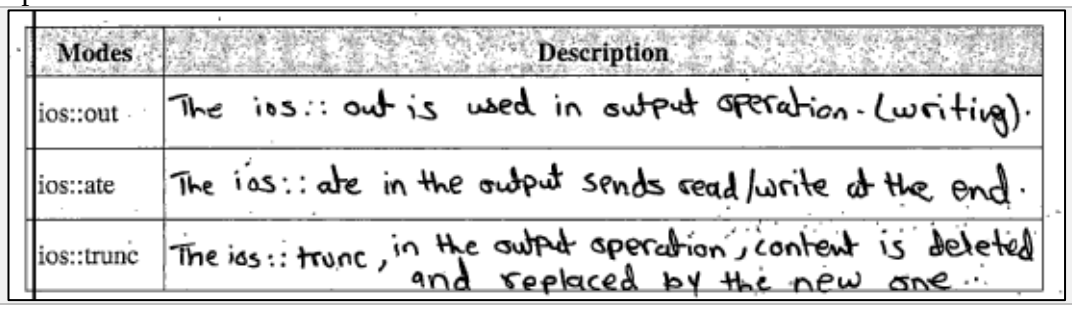
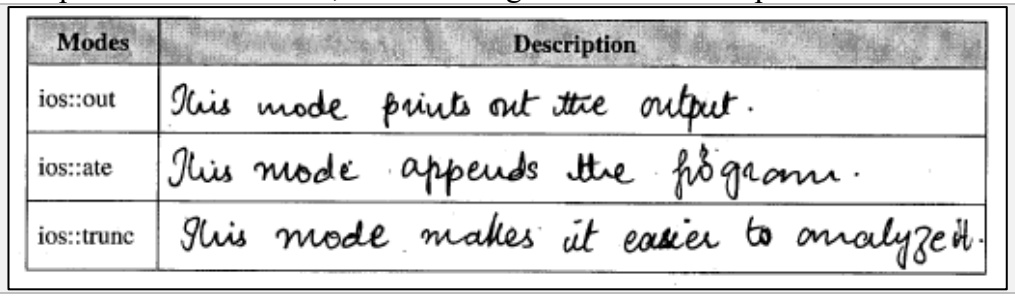
Image of Weaker Response	

```
#include <iostream>
using namespace std;
class Animal {
  public:
   void eat() {
      cout << "I can eat!" << endl;
   }
   void sleep() {
      cout << "I can sleep!" << endl;
   }
};
class Dog : public Animal {
  public:
   void bark() {
      cout << "I can bark! Woof woof!!" << endl;
   }
};
int main() {
   // Create object of the Dog class
      Dog, B1;
 // Calling members of the base class
      show data.eat();
      show data.sleep();
// Calling member of the derived class
      show data.B1;
return 0;
}
```

Suggestions for improvement (Tick all that apply)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept mapping • Audio Visual resources • Think, pair and share • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform
Any Additional Suggestion: Teachers are recommended to give more practice problems involving the concept of inheritance and other pillars of Object-Oriented Programming (OOP) using C++.		

Question No. 6	
Question Text	Describe each of the given modes of opening a file.

	<table><tr><th>Modes</th><th>Description</th></tr><tr><td>ios::out</td><td></td></tr><tr><td>ios::ate</td><td></td></tr><tr><td>ios::trunc</td><td></td></tr></table>
SLO No.	17.1.2
SLO Text	Compare different modes of opening a file.
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for writing each description.
Overall Performance	The question addresses various methods of file opening in C++. In aggregate, the cohort's responses can be deemed satisfactory. A majority of candidates displayed accurate understanding by providing correct answers, reflecting competence in the topic. Nonetheless, some responses were less robust, implying insufficient practice in C++ file handling. Encouragement is extended for continued engagement and practice in mastering file handling techniques for enhanced proficiency across the cohort.
Description of Better Responses	The better responses correctly showcased comprehension of file modes, providing a coherent description with precise details. The explanation effectively expounded on the functionality of file modes, underscoring a robust understanding of the subject matter. Additionally, the responses correctly distinguished modes for input, append, and output operations.
Image of Better Response	
Description of Weaker Responses	Weaker responses showcased limited comprehension of file modes in C++, incorporating inaccuracies and unrelated elements. Mostly candidates wrote general description against each file mode like ios::out is used to print the output or ios:: app is used to append the program. These aspects detracted from the topic's clarity and demonstrated weaker understanding. To enhance performance, the candidates are advised to focus on accurate and pertinent information, concentrating on the core concepts of file modes in C++.
Image of Weaker Response	

Suggestions for improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
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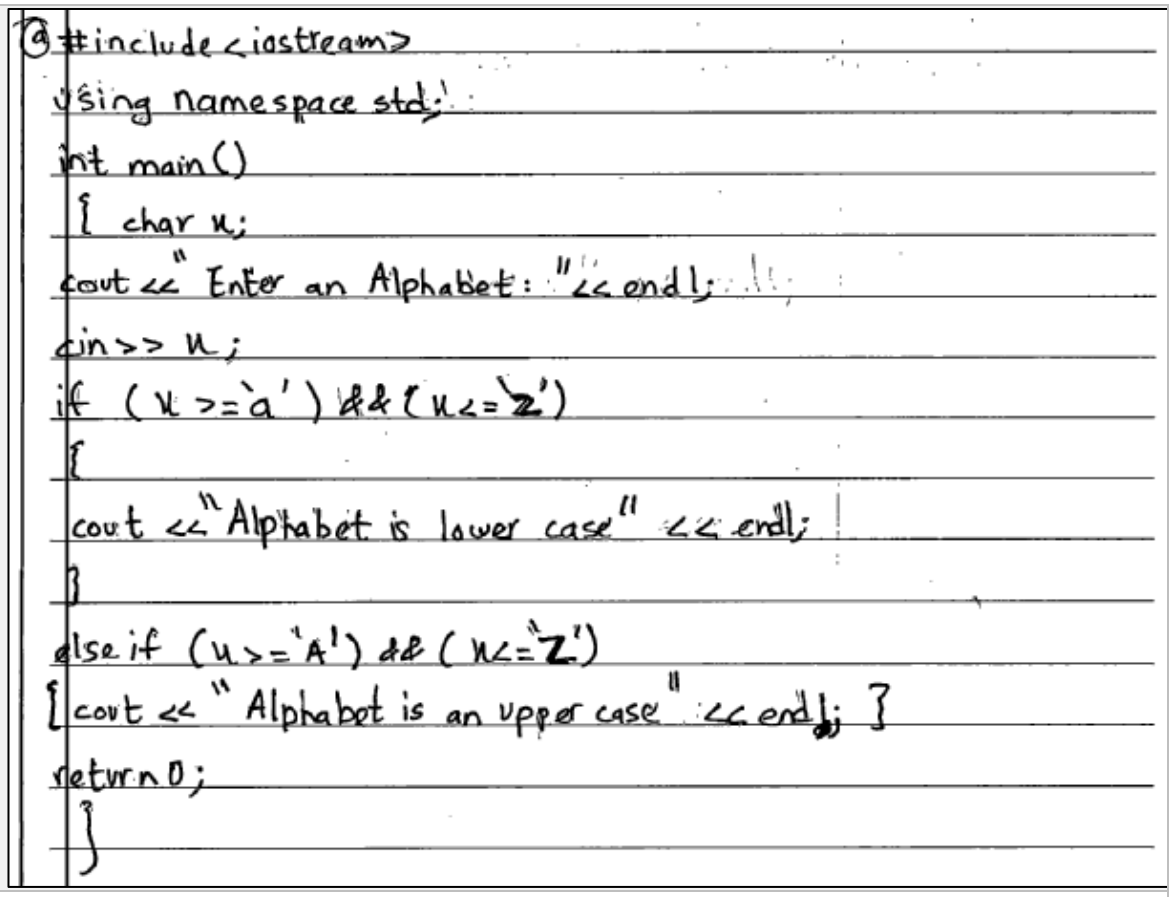
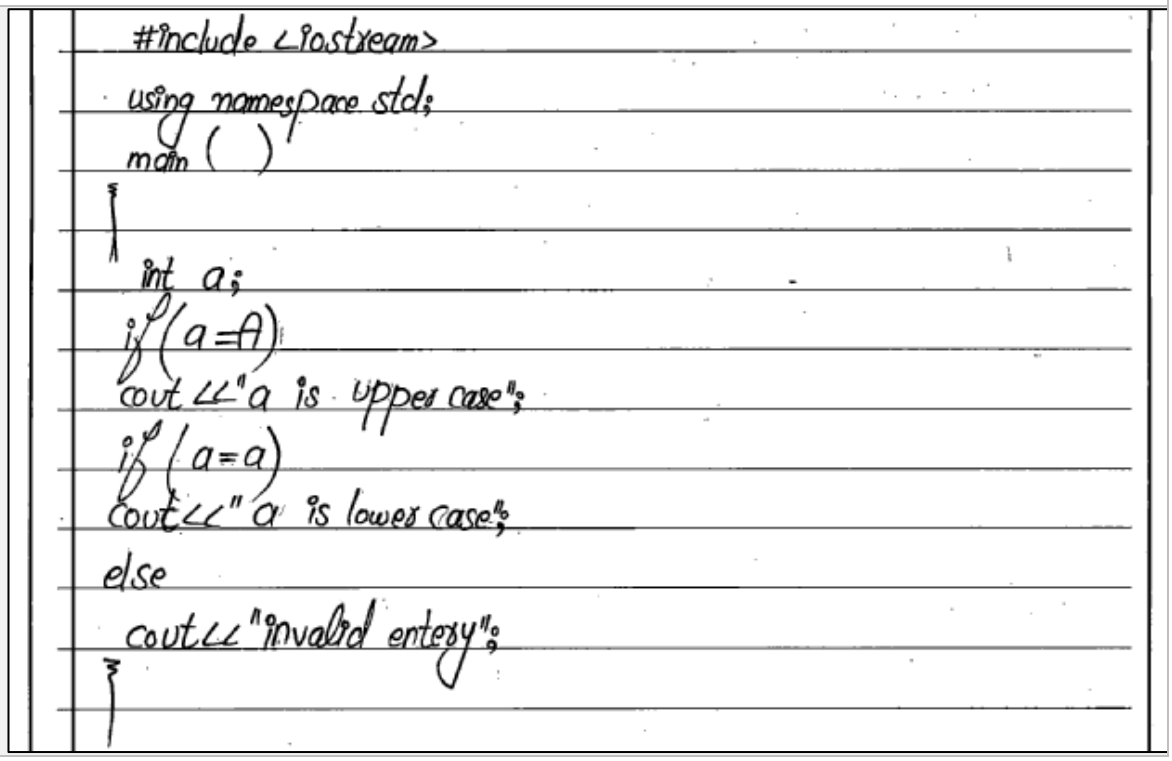
• Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Audio Visual resources • Think, pair and share • Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion:
Teachers are recommended to give more practical problems such as writing, deleting or append data into the file.

Extended Response Questions (ERQs)

These questions offered a choice between part **a** and **b**.

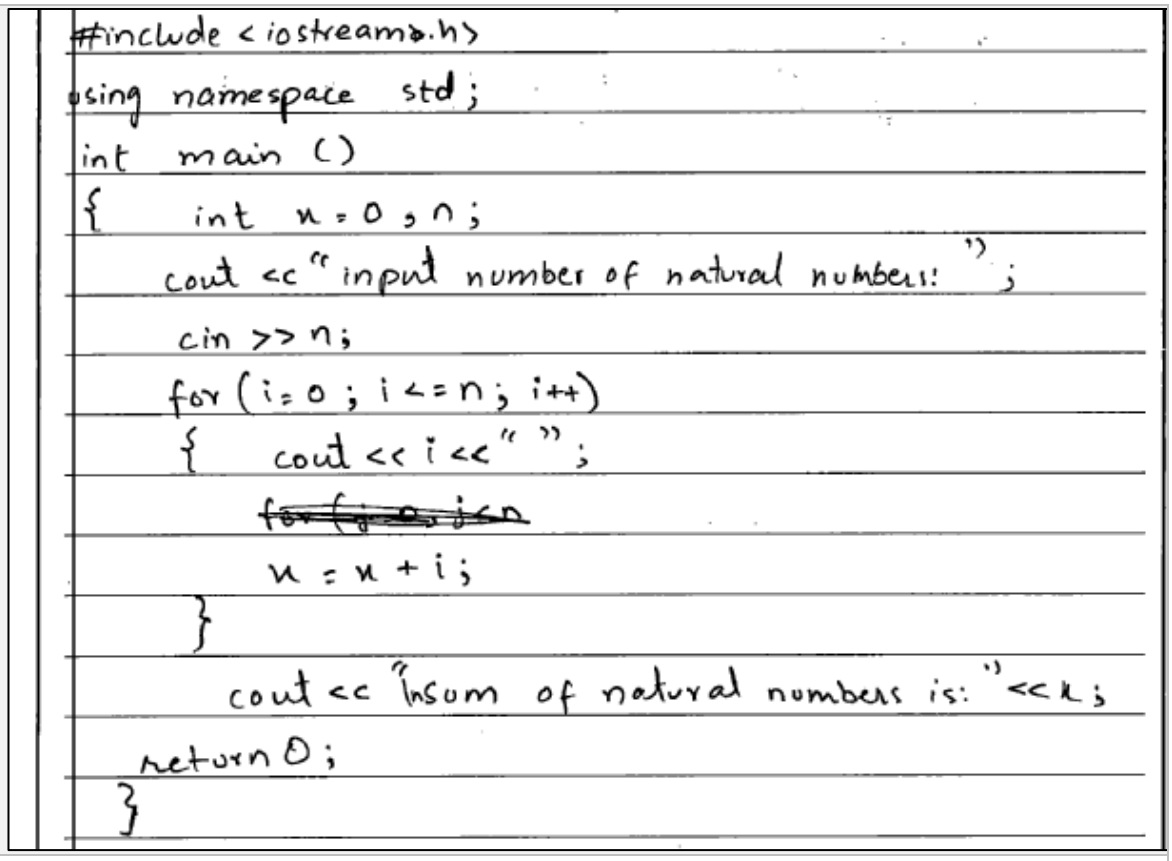
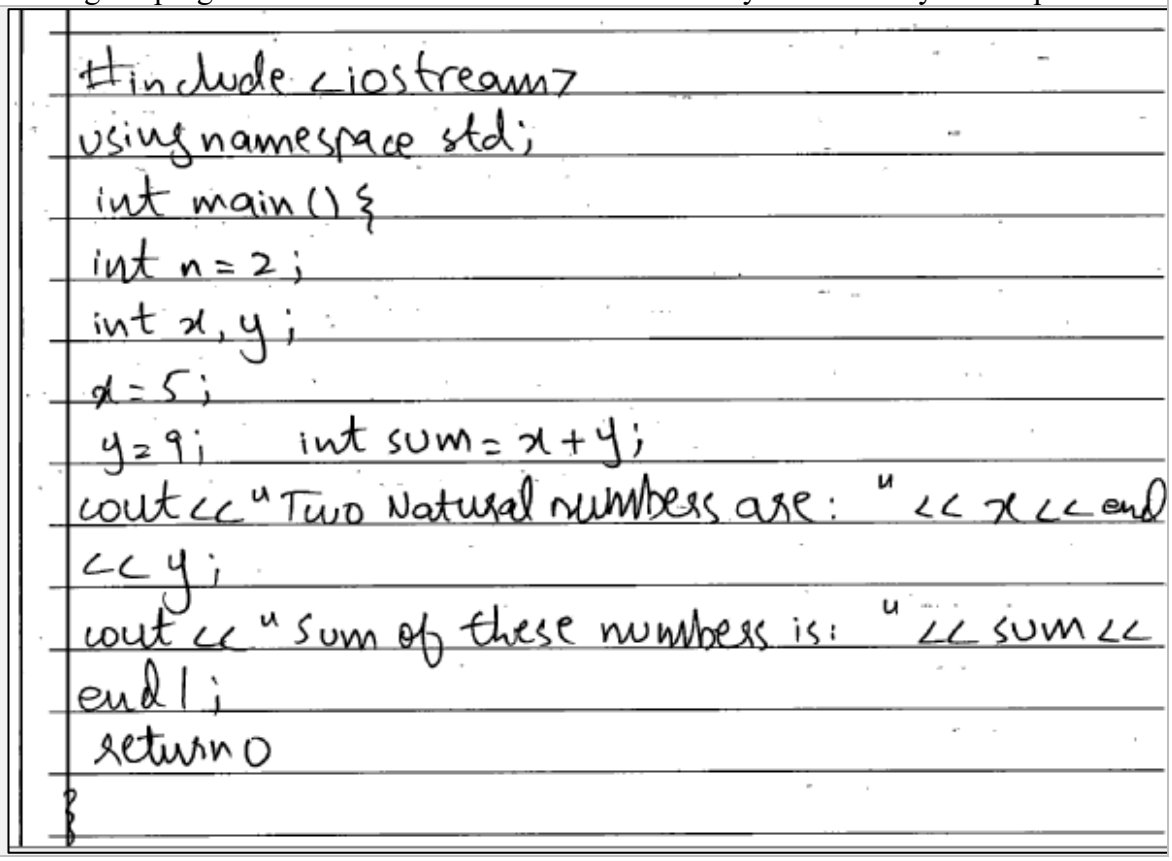
Question No. 7a	
Question Text	Write a C++ program to check whether an inputted alphabet is upper case or lower case.
SLO No.	11.4.8 12.1.2
SLO Text	Write a program using relational operators. Write a program for each **if**, **if-else** and **else-if** statements.
Max Marks	7
Cognitive Level	A
Checking Hints	1 mark for declaring variable 1 mark for taking input 1 mark for writing if structures (TWO required) 1 mark for using relational operator in one condition (TWO required) 1 mark for writing output
Overall Performance	As this was an ERQ, almost half of the candidates attempted this part of the question. This question was about writing C++ code to figure out whether input letters are in uppercase or lowercase. Responses exhibited a spectrum of comprehension, with some showcasing adeptness in program logic and structure. These responses featured accurate variable declarations, inputs, outputs, and conditional statements. Nonetheless, some of the responses suggested a less conceptual understanding, potentially falling short of meeting the question's requisites.
Description of Better Responses	Better responses effectively displayed the grasp of the task, proficiently employing accurate logic to devise a program. The inclusion of precise input, output, logical operators and conditional statements reflected a sound understanding of programming principles. Notably, the responses showcased suitable application of correct syntax, underscoring the candidates' command over the programming language and logical constructs. In totality, such answers exemplified a well-executed approach, effectively addressing the question's stipulations.

Image of Better Response	
Description of Weaker Responses	The candidates' responses exhibited deficiencies, marked by erroneous logic and inaccuracies within the statements. Using conditional statements and logical operator incorrectly underscore a lack of clarity in comprehension, resulting in a less proficient execution.
Image of Weaker Response	

Suggestions for improvement (Tick all that apply)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept mapping • Audio Visual resources • Think, pair and share • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform
Any Additional Suggestion: Teachers are advised to give more practical problems such as https://clas.iusb.edu/docs/CSCI/C101-PracticeProblems.pdf and ask candidates to practice this concept.		

Question No. 7b	
Question Text	Write a C++ program to display n terms of natural numbers and their sum.
SLO No.	11.4.13 12.2.2
SLO Text	Write a C++ program that uses compound expression. Write a C++ program that uses **for** loop.
Max Marks	7
Cognitive Level	A
Checking Hints	1 mark for declaring variables 1 mark for taking input 1 mark for writing for loop structure 1 mark for writing condition of for loop 1 mark for writing n terms 1 mark for the formula to sum natural numbers 1 mark for writing the sum of natural numbers
Overall Performance	As this was an ERQ, almost half of the candidates attempted this part of the question. The overall performance of the candidates in this question was commendable. Noteworthy responses encompassed accurate variable declarations, logical precision, and a coherent program structure. Predominant responses demonstrated correct and executable logic. Nonetheless, a subset of candidates exhibited misconceptions and encountered challenges in program logic, signaling an opportunity for heightened practice and consolidation in this domain.
Description of Better Responses	Better responses reflected a strong grasp of the task's essence, displaying precise comprehension. These candidates accurately declared variables, deployed correct logic, used the 'for loop' correctly and structured the program adeptly. Notably, the counter statements' placement for sum calculation underscored adept logic handling, with accurate parameter positioning. the overall execution underscored a robust command of core concepts, proficiently addressing the question's requirements.

Image of Better Response	
Description of Weaker Responses	Weaker responses revealed a constrained understanding of the concepts, evident through erroneous variable initialisation, absence of loop structure, and flawed programming logic. Addressing these deficiencies and incorporating accurate variable setup, loop utilisation, and logical progression would enhance the overall efficacy and accuracy of the question.
Image of Weaker Response	

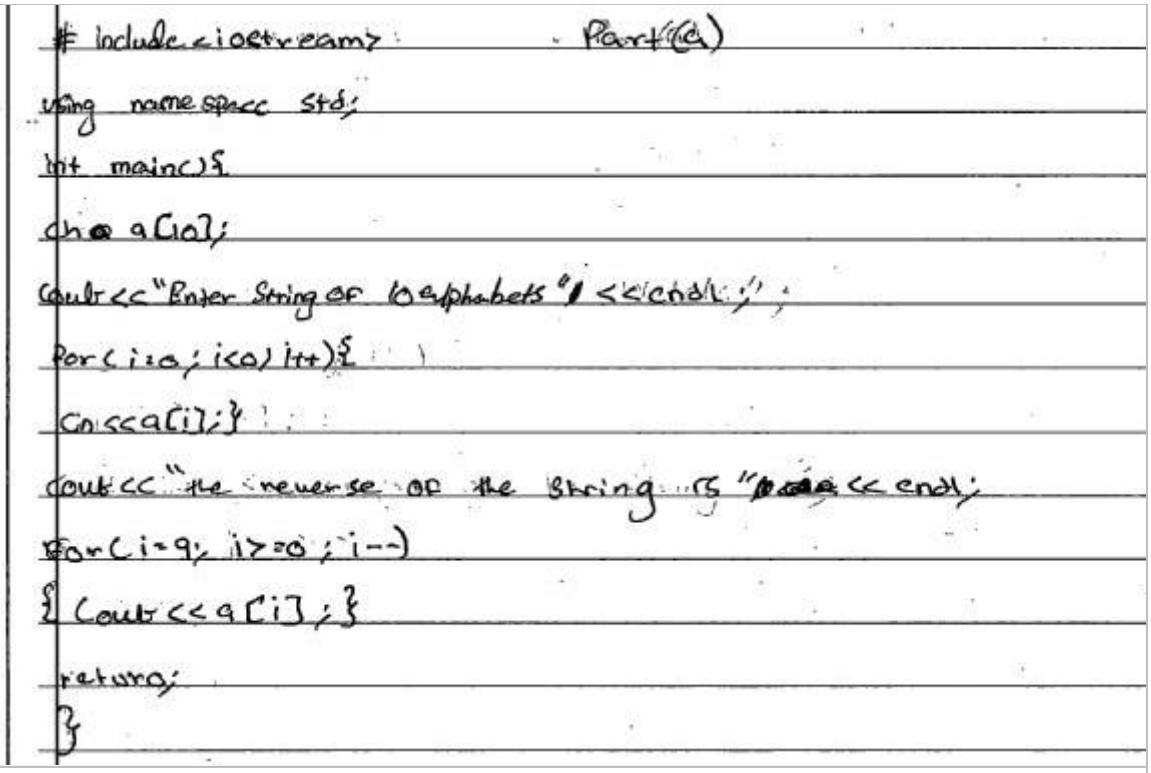
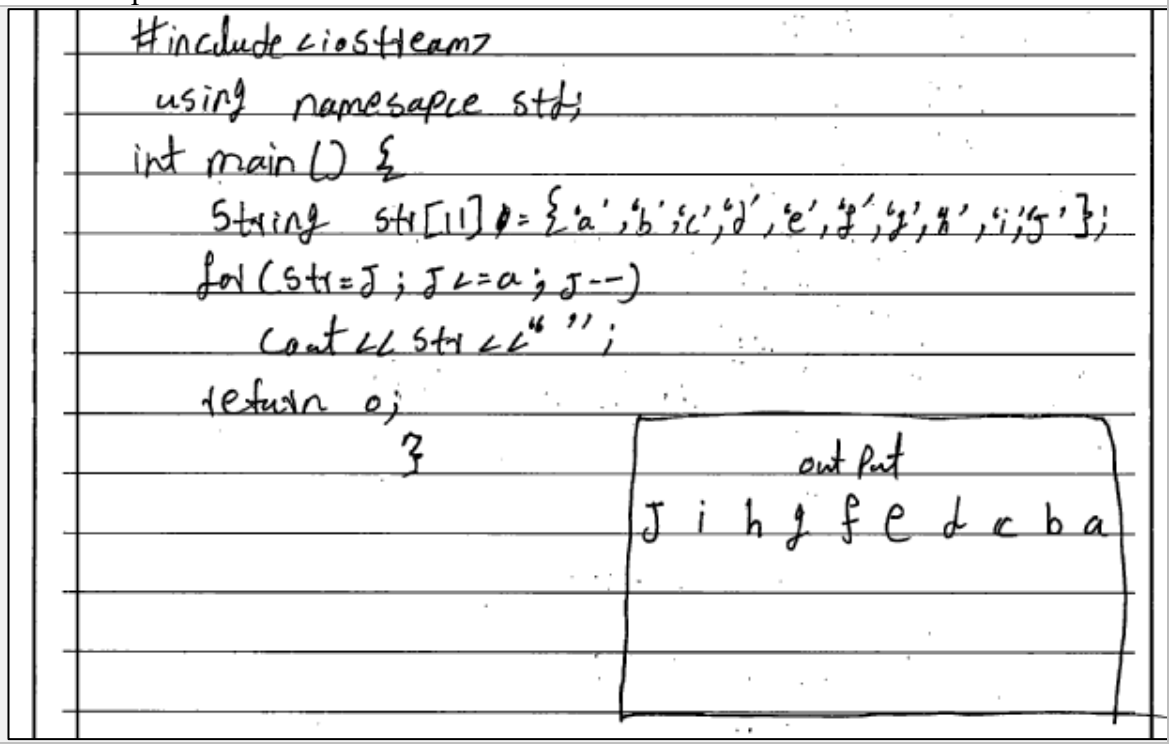
```
#include <iostream.h>
using namespace std;
int main ()
{   int x = 0 , n;
    cout << " input number of natural numbers: ";
    cin >> n;
    for ( i = 0 ; i <= n; i++)
    {   cout << i << " ";
        for (j = 0; j<n
        x = x + i;
    }
      cout << "\nsum of natural numbers is: " << x;
  return 0;
}
```

```
#include <iostream>
using namespace std;
int main(){
int n = 2;
int x, y;
x = 5;
y = 9;   int sum = x + y;
cout << "Two Natural numbers are: " << x << endl
<< y;
cout << "sum of these numbers is: " << sum <<
endl;
return 0
}
```

Suggestions for improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept mapping • Audio Visual resources • Think, pair and share • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform
Any Additional Suggestion: Teachers are advised to give more practical problems such as https://clas.iusb.edu/docs/CSCI/C101-PracticeProblems.pdf to the candidates for the better understanding of the concept.		

Question No. 8a	
Question Text	Write a C++ program that reverses an inputted string of 10 alphabets and prints it on the screen.
SLO No.	13.1.7
SLO Text	Write a program which stores numeric values in a one-dimensional array using **for** loop and finds the highest, lowest and average values.
Max Marks	7
Cognitive Level	A
Checking Hints	1 mark for declaring variables (TWO required) 1 mark for taking input 1 mark for using for loop 1 mark for placing values in reversed order 1 mark for using for loop showing output 1 mark for writing the reversed string
Overall Performance	As this was an ERQ, few candidates attempted this part of the question. The candidates' overall performance in this question was below average. The responses displayed deficiencies in proper programming structure, including incorrect statement placement and variable declaration. While a few responses showed better answers, none were entirely accurate. Enhanced understanding of variable declaration, loop structures, and programming statements can be achieved through more practice of similar questions, enabling candidates to refine their programming skills and solidify their conceptual grasp.
Description of Better Responses	Even though the candidates' better responses did not attain complete accuracy, they did exhibit an enhanced comprehension of programming logic. Notably, candidates displayed improved understanding in aspects such as variable declaration, initialisation, and utilisation of loop structures. This indicated a positive trend towards refining programming skills and grasping fundamental concepts.

Image of Better Response	
Description of Weaker Responses	The weaker responses demonstrated weak logic, incorrect declaration and initialisation of variables, use of loop structure and output statement lead to inaccuracies in outcomes. Improving the logical flow and ensuring accurate input and output statements with correct use of loop structure would enhance the correctness of the solution.
Image of Weaker Response	

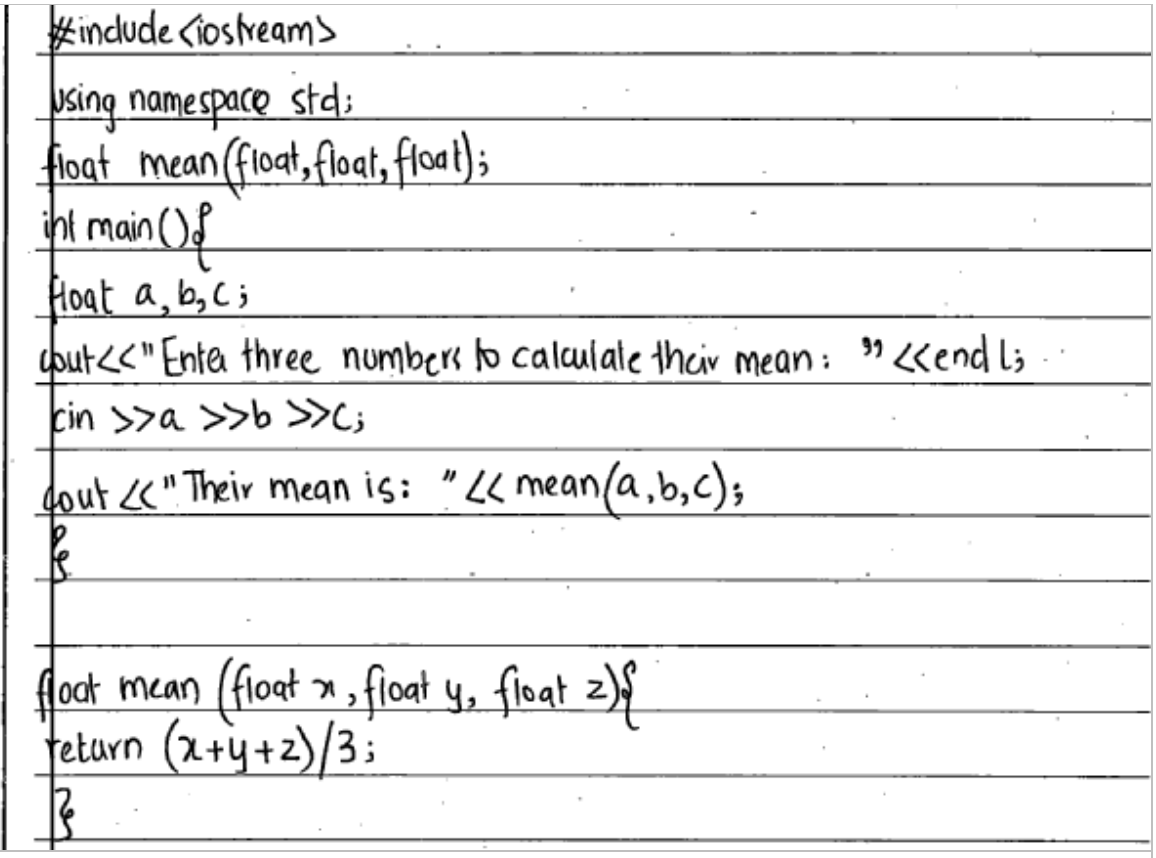
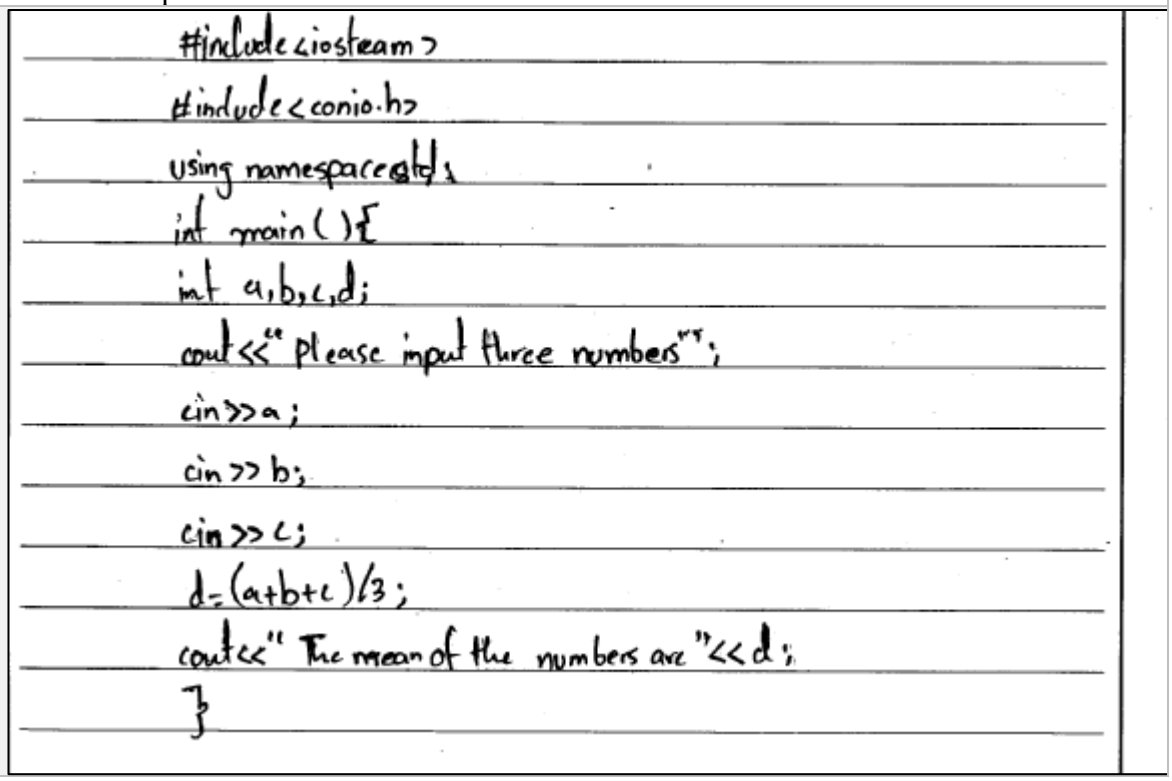
Suggestions for improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and	• Story Board • Cause and Effect • Fish and Bone • Concept mapping • Audio Visual resources • Think, pair and share	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform

any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion:
Teachers are advised to provide the online resources of different programming problems such as https://www.codesdope.com/practice/cpp-array/ for more practice and the better understanding of the concepts.

Question No. 8b	
Question Text	Write a C++ program using user defined function that takes three numbers as an input and print their mean.
SLO No.	14.2.1
SLO Text	Write a program to invoke a user defined function and pass arguments by constants, value and reference;
Max Marks	7
Cognitive Level	A
Checking Hints	1 mark for writing function prototype 1 mark for declaring variables in main function 1 mark for taking input in main function 1 mark for calling function 1 mark for writing correct parameters 1 mark for writing function definition 1 mark for writing output
Overall Performance	As this was an ERQ, many candidates opted to attempt this part of the question. In addressing this question, the candidates' responses were notably above average. Many effectively wrote programs with accurate logic, demonstrating a solid understanding of the task. However, some weaker responses lacked a crucial element, the inclusion of a user-defined function which is pivotal to the question's requirements. This omission impacted the completeness of the solution and highlights an area for improvement in those specific responses.
Description of Better Responses	The candidates' responses demonstrated an exceptional understanding of programming concepts, featuring a coherent and organised logic. The inclusion of a well-structured user-defined function with accurate data types and its proper invocation in the main function reflected a solid grasp of program design. The candidates clearly applied appropriate program structure, showcasing a good command of coding principles and practices.

Image of Better Response	
Description of Weaker Responses	The candidates' responses lacked a user-defined function to calculate the mean of three numbers, resulting in incomplete program code. This omission limits the program's functionality and leads to reduced marks for input and output segments. Encouraging candidates to make programs using user defined function can lead to better understanding of the concept.
Image of Weaker Response	

Suggestions for improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept mapping • Audio Visual resources • Think, pair and share • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform
Any Additional Suggestion: Teachers are advised to give the different programming problems of user defined functions such as https://www.codesdope.com/practice/cpp-function/ for the better understanding of the concepts.		

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard

Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.

Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect

Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.

Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone

Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.

Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping

Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.

Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources

Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.

Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share

Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.

Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)

Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.

Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration

Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.

Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

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